Ondřej Baštař

bastarondrej@outlook.com ◇ +420 605 083 960 ◇ Prague

WORK EXPERIENCE

Vrgineers *Mixed Reality researcher/developer* **2023 – Present**

- Lead developer on the mixed reality pipeline for the XTAL/somnium VR1 headset
- Development of a real-time c++/cuda image processing pipeline which handles 5000x5000@102fps images through stages (demosaicking, denoise, sharpening, distortion, …).
- Research work in optimizing the various image processing stages e.g. improving the denoiser from being a major performance hit to being negligible at the same qualitative performance, via low-level cuda optimizations

FEE CTU *Teaching Assistant* **2025 - Present**

- I teach tutorials for first-year Linear Algebra for the open informatics students.

AI Center (FEE CTU) *NLP Researcher* **2022 – 2023**

- University research internship. I focused on sentence-to-sentence similarity and building datasets
- Built a sentence similarity dataset leveraging cross-source news coverage of the same events to generate naturally similar sentence pairs.

COMTES FHT *Engineer* **2020 – 2022**

- I developed solutions to increase productivity and efficiency of researchers:
  - Software and hardware for a prototype of an affordable metal 3D printer (modified Průša printer)
  - IoT monitoring solution for industrial machinery based on Arduino and Raspberry PIs.
  - GUI program for training and using neural networks to analyze images in Imgui and Python

Whitesoft *Freelance IoT Engineer* **Summer 2020**

- Developed a smart public announcement system for a retirement home.
  - Backend in flask with a website in Vue.js.
  - Custom modular solution for Raspberry PIs built in Python.

EDUCATION

Czech Technical University in Prague **2022 – Present**

Faculty of Electrical Engineering - Cybernetics and Robotics *Prague*

Extra courses: Functional analysis, 4 semesters of phd theoretical physics. RTOS programming. Various advanced control algorithms.

Thesis (bachelor): Using magnetohydrodynamics (MHD) to improve cooling properties of a conductive liquid.

Supervisor: Ing. Jiří Zemnáek, Ph. D.

- Development of a custom MHD solver for the OpenFOAM system.
- Real Time MPC control using the simulations and surrogate models.
- Building the hardware for verifying control/simulation using existing solenoid boards from previous work of a researcher at the university.

COMPETITIONS

1st place in Microsoft AI accessibility hackathon (AiAiEyes)

- ~15 teams both industry and students

- We developed the application AiAiEyes, unique visual aid for vision-impaired (see Extracurricular)

1st place in international Vrgineers Inc. Algorithm design competition (university news report link)
- Implemented image processing algorithms for real-time stereo 4k stream using compute shaders and CUDA, achieving processing in a few milliseconds on an RTX 2080
- Competed against 25 international competitors

1st place in national Porsche Engineering Student competition (10-15 student teams)
- Built a racing algorithm achieving the fastest lap around a racetrack in the CARLA simulation, taking camera stream as input and outputting throttle and steering

1st place in the national LIFMAT AI 2024 hackathon (AI in healthcare)
- 20 teams both students and industry
- Developed a solution for digitalizing and summarizing medical reports, because in Czechia they still exist only on paper.

1st place in the national round of EuroTeQaThon (15 student teams from Czechia)
- Audio processing solution for detecting malfunctions in Asahi breweries in Czechia. We used a mix of a hardware IoT solution and audio/signal processing (with vector embeddings) which would detect anomalous sounds in a plant, detecting mechanical failure early.

EXTRACURRICULAR

TePrVe (Technical evening of presentations teprve.org, press release) (czech version only)
- Started this student event, where we meet up (currently about 40 attendees) and give each other lectures/talks about our technical projects or stuff that interests us.

AiAiEyes
- AI based visual aid application for visually-impaired people. In development with SONS (czech association of the visually impaired).
- Enables users to locally (on smartphones) scan their objects and create a database of things. They can then use this database to recognize items from each other. For example T-Shirt with differing colors or grocery labels.

Talks
- **How to backpropagate through fluids**: (TePrVe čtvrtý: link in czech) a talk on differentiable fluid sims in nvidia-warp
- **CUDY z nudy:** (TePrVe první: link in czech) introductory talk on cuda, but the pun doesn't work in english
- **How to get study notes from slides?:** (on 4.3.2026, no link yet) a project for using an agentic LLM harness to convert sparse slides into fluid and well readable notes.
- **LectureRAG:** (AI Tinkerers: link): A **talk on** creating a knowledge base from lecture recordings
- **Pinakes:** (AI Tinkerers: link): Experimental tool for evaluating how well can agents keep notes organized

SKILLS & INTERESTS

- **Skills:** Python, Javascript, C, SQL, PyTorch, Linux, AWS, 3D printing, GPU Programming
- **Interests:** 3D printing, Stock Trading, Linux, Experimenting with LLMs, Futurama
- **Languages:** Czech, fluent English